

Maël Boutsoque

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Resume

As a Digital Systems Engineering student at **ENSEM**, I am soon to graduate and am actively seeking a job in hardware. My academic background and interest in embedded systems and electronics have strengthened my motivation to develop practical skills in low-level programming, hardware design, and real-time systems.

Formation

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|---|----------------|
| ENSEM , Digital/embedded systems engineer | 2023 - present |
| <ul style="list-style-type: none">• Courses: System control, Optimization, Computing, Electronics, Real-time systems, Data analysis, System safety. | |
| Preparatory class for entrance to French engineering schools (CPGE PC) | 2021 - 2023 |
| <ul style="list-style-type: none">• Courses: Mathematics, Physics, Chemistry, Computer Science, Engineering Science. | |

Experience

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|--|----------------|
| QUANDELA : Hardware/Software Designer (6 month internship) | 2026 - present |
| Projects : | |
| <ul style="list-style-type: none">• Communication protocol migration of 2 proprietary protocols:<ul style="list-style-type: none">– QDL : RS485 based, migrated from AVR to STM32 using standards C libraries– QDL fast : Ethernet based, migrated from linux to STM32 using Hal library and FreeRTOS• Design of multifunction Hub PCB board in Altium<ul style="list-style-type: none">– Setting up a specification document based on the requirements– Component selection based on the specifications and datasheets– Component placement and routing | |
| Board Functions : Ethernet Hub, RS485 Hub, I2C modules, USB flash, Multi-function Relays, Servo Motors, ... | |

Skills

Computer languages :

low level : C C++ assembly | Arduino Hal Bare metal RTOS
software : Python Java Matlab | Git CI/CD CMake

Tools :

apps Microsoft Office Suite | Latex | VScode Eclipse STM32 suite
PCB Altium Kicad | Fusion 360 | Eurocircuit
data SQL Json | AI dataset AI training

Languages :

- **French** Native
- **English** B2+ (920 Toeic)
- **Spanish** B1

Projects

Arduino system design and development (*Personal project*) 2022 - present

- Several personal projects involving the creation of electronic systems (e.g. real-time radar, remote-controlled car, robot arm).
- Tools : Arduino Framework

ENSEM Eco Marathon (*Group project*) 2023 - 2026

- Design and manufacture of a test bench to measure the performance of a vehicle in order to improve its performance under the constraints of the standards of a race organized by Shell.
- Secondary car driver

Autonomous robot (*Group project*) 2023 - 2024

- Control of an autonomous geolocation robot capable of autonomously and optimally navigating to a list of coordinates.
- Tools : Python, TCP protocols

Video game development (*Personal and Group projects*) 2023 - 2025

- Design 2D video games using basic graphical user interface libraries in Java and Python.
- Tools : Python, Java